



Chao Xu

Date and place of birth: 21st Nov. 1996, Shandong, China

Nationality: Chinese

Language: Chinese, English, French (basic)

Lab: Center for Combinatorics 308, Nankai University

+86-13626410052

XClon1@gmail.com

Chao_xu@nankai.edu.cn

EMPLOYMENT

• Nankai University, Tianjin

09/2026-06/2029

Postdoc in Center for Combinatorics, Advisor: Dun Qiu and Arthur Li Bo Yang

EDUCATION

• Jinan University, Jinan Shandong

09/2015-06/2019

B.S. in Financial Mathematics.

• Shandong University, Qingdao Shandong

09/2019-06/2022

M.S. in Science. Advisor: Zhicong Lin

• Université Claude Bernard Lyon 1, Lyon France

09/2022-06/2026

Ph.D. in Mathematics. Advisor: Jiang Zeng

ACADEMIC SERVICE

•Reviewer for Mathematical Reviews, since 2026

RESEARCH INTEREST

- **Enumerative combinatorics:** Symmetric functions, Permutation statistics, Orthogonal polynomials and Continued fractions, ASEP

RESEARCH

• Master thesis

Permutation Statistics and Chen's Context-free Grammars

- Applications of Chens context-free grammars in permutation statistics
- Combinatorial interpretation of the gamma-coefficients of multiset Eulerian polynomials

• PhD thesis

Combinatorics of (p, q) -analogues of bi-Stirling Eulerian polynomials

- Generalizations of Eulerian polynomials and their gamma-positivity
- Jacobi-type continued fractions and (p, q) -analogs of generalized Eulerian polynomials
- Combinatorial interpretations of remixed Eulerian numbers
- Enumeration of colored permutations by the parity of descent positions

• Publications and Preprints

- On the γ -positivity of multiset Eulerian polynomials, with Zhicong Lin and Tongyuan Zhao, *European Journal of Combinatorics* **102** (2022), Paper No. 103491, 18 pp. [Journal](#)
- Enumeration of colored permutations by the parity of descent positions, with Qiongqiong Pan and Jiang Zeng, *Enumerative Combinatorics and Applications* **4** (2024), no. 3, Paper No. S2R22, 15 pp. [Journal](#)
- Variations of the (α, t) -Eulerian polynomials and γ -positivity, with Jiang Zeng, in *Proceedings of the 37th Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2025, Sapporo)*, *Séminaire Lotharingien de Combinatoire* 93B (2025), Article 41, 12 pp. [Proceedings](#)
- A bi-Stirling–Euler–Mahonian polynomial, with Jiang Zeng, *Advances in Applied Mathematics* (2026) **173**, 102975. [Journal](#)
- Variations of (α, t) -Eulerian polynomials and their gamma positivity, with Jiang Zeng, to appear in *Combinatorial Theory* [arXiv](#)
- Combinatorics of (p, q) -analogues of bi-Stirling Eulerian Polynomials, with Jiang Zeng, submitted.
- A Refinement of the Fixed–Fixed Points Equidistribution on restricted Permutations, with Yao Dong, submitted, [arXiv](#)
- A combinatorial interpretation of Désarménien's formula, submitted.
- New Laguerre History Interpretations of Euler Numbers and Reduced Tangent Numbers, submitted.
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CONFERENCES ATTENDED

- The 11th Cross-Strait Conference on Graph Theory and Combinatorics, Aug. 20–22, 2021 (gave a talk)
- The 12th Computer Mathematics Conference (CM 2021), Jun. 4–7, 2021
- The 91th Séminaire Lotharingien de Combinatoire, Germany, Sep. 3–6, 2023 (gave a talk)
- The 92th Séminaire Lotharingien de Combinatoire, Spain, Mar. 17–20, 2024
- Journées du GT Combinatoire algébrique, Lyon (ICJ), Oct. 14–16, 2024
- Congrès des Jeunes Chercheurs en Mathématiques Appliquées, Lyon (ENS), Oct. 28–30, 2024
- Conférence de clôture ANR Combiné, Lyon (ICJ), Mar. 10–11, 2025 (gave a talk)
- The 37th Formal power series and Algebraic combinatorics, Japan, Jul. 21–25, 2025 (poster)
- International Conference on Enumerative Combinatorics and Applications 2025 (on line, Haifa), August 25-27, 2025 (gave a talk)
- Journées ALEA, Marseille, France, Mar. 9–13, 2026
- The 95th Séminaire Lotharingien de Combinatoire, Lyon, France, Mar. 22–25, 2026

HONORS AND REWARDS

- Awarded a grant from China Scholarship Council (CSC) for PhD study at Université de Lyon for 4 years

SKILLS

- **Software:** Microsoft Word, Excel, Mathematica, Python, Latex, SageMath

The essence of mathematics is its freedom.